

National University of Computer and Emerging Sciences

Assignment 02

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MT1003 Calculus and Analytical Geometry
BS CS

Due Date: Sunday, October 8, 2023, 11:59PM.

Differentiation

Q. 1 City Revenue. The revenue realized by a small city from the collection of fines from parking tickets is given by

$$R(n) = \frac{8000n}{n+2},$$

where n is the number of work-hours each day that can be devoted to parking patrol. At the outbreak of a flu epidemic, 30 work-hours are used daily in parking patrol, but during the epidemic that number is decreasing at the rate of 6 work-hours per day. How fast is revenue from parking fines decreasing at the outbreak of the epidemic?

Q. 2 Compound Interest. Suppose a sum of \$500 is deposited in an account with an interest rate of r percent per year compounded monthly. At the end of 10 years, the balance in the account (as illustrated in Figure 10) is given by

$$A = 500 \left(1 + \frac{r}{1200} \right)^{120}.$$

Find the rate of change of A with respect to r if $r = 5$ or 7 .*

[* Notice that r is given here as an integer percent, rather than as a decimal, which is why the formula for compound interest has 1200 where you would expect to see 12. This leads to a simpler interpretation of the derivative.]

Q. 3 Suppose that $u(t)$ measures the displacement (measured in inches) of a weight suspended from a spring t seconds after it is released and that $u(t) = 4 \cos t$. Find the velocity at any time t and determine the maximum velocity.

Q. 4 Tangent Line

- Use implicit differentiation to find dy/dx for the Folium of Descartes $x^3 + y^3 = 3xy$.
- Find an equation for the tangent line to the Folium of Descartes at the point $\left(\frac{3}{2}, \frac{3}{2}\right)$.
- At what point(s) in the first quadrant is the tangent line to the Folium of Descartes horizontal?

Q. 5 Implicit Differentiation

- Find x' for $x = x(t)$ defined implicitly by $t \ln x = xe^t - 1$ and evaluate x' at $(t, x) = (0, 1)$.
- Find x' for $x = x(t)$ defined implicitly by $1 + x \ln t = te^x$ and evaluate x' at $(t, x) = (1, 0)$.

Q. 6 Logarithmic Differentiation

Differentiate the following functions by logarithmic differentiation.

a.

$$y = \frac{x^{3/4} \sqrt{x^2 + 1}}{(3x + 2)^5}.$$

b.

$$y = x^{\sqrt{x}}.$$

c.

$$y = (x^2 + 1)^{\sin x}.$$

Q. 7 A point P moves along the x –axis in such a way that its position at time t s is given by

$$x = 2t^3 - 15t^2 + 24t \text{ ft.}$$

- a. Find the velocity and acceleration of P at time t .
- b. In which direction and how fast is P moving at $t = 2$ s? Is it speeding up or slowing down at that time?
- c. When is P instantaneously at rest? When is its speed instantaneously not changing?
- d. When is P moving to the left? to the right?
- e. When is P speeding up? slowing down?

Q. 8 A Sawtooth Curve. Let $f(x) = \sin^{-1}(\sin x)$ for all real numbers x .

- a. Calculate and simplify $f'(x)$.
- b. Where is f differentiable? Where is f continuous?
- c. Use your results from (a) and (b) to sketch the graph of f .
- d. Plot the graph of f and f' using computing tool.

Q. 9 Differentiation of Inverse Function(s)

- a. Show that $f(x) = x^3 + x$ is one-to-one on the whole real line, and, noting that $f(2) = 10$, find $(f^{-1})'(10)$.
- b. Evaluate $f'(2\sqrt{3}) = x \tan^{-1}(x/2)$.
- c. Find the derivative of $\sin^{-1}\left(\frac{x}{a}\right)$ and hence evaluate $\int \frac{dx}{\sqrt{a^2 - x^2}}$ where $a > 0$.
- d. Find the derivative of $\tan^{-1}\left(\frac{x}{a}\right)$ and hence evaluate $\int \frac{dx}{x^2 + a^2}$.

Q. 10 (MATLAB)

A lighthouse L is located on a small island 2 km from the nearest point A on a long, straight shoreline. If the lighthouse lamp rotates at 3 revolutions per minute, how fast is the illuminated spot P on the shoreline moving along the shoreline when it is 4 km from A ?

Write a MATLAB code for the problem that works for given numerical inputs and produce desired results as per your formulation for the problem.

Good Luck